arXiv:2610.13916v1 [math.PR] 9 Oct 2026

# Markov Chain Convergence in Kernel-Induced Geometry

Oleksii Kachaiev[1], Luc Brogat-Motte[2], Silvia Villa[1], Lorenzo Rosasco[2,3]

[1]MaLGa center, DIMA, Università degli Studi di Genova, Genoa, Italy

[2]SAIL Unit, Istituto Italiano di Tecnologia, Genoa, Italy

[3]MaLGa Center, DIBRIS, Università degli Studi di Genova, Genoa, Italy

Correspondence to: `oleksii.kachaiev@edu.unige.it`

**Abstract**

We study the convergence of Markov chains in the geometry induced by Hilbertian embeddings associated with positive definite kernels, also called Maximum Mean Discrepancy (MMD). Under a contraction assumption in the kernel-induced geometry, and a standard spectral gap assumption, we establish non-asymptotic, high-probability bounds for the convergence of the empirical occupation measure to the invariant measure. The obtained bounds separate the roles of the two conditions: the spectral gap determines the effective sample size in the leading term, while the kernel contraction enters only through a linear correction. As an application, we study one-step prediction of dynamical systems from a single trajectory, deriving fast learning rates for a nonlinear least-squares estimator. The rates highlight the interplay between mixing properties in the kernel-induced geometry and the classical source and capacity conditions used in the analysis of least-squares.

## 1 Introduction

Let $(X_t)_{t\geq 0}$ be a Markov chain on $\mathfrak{X}$ with transition kernel $P$. Its *empirical occupation measure* up to time $T$ is

$$\widehat{\mu}_T := \frac{1}{T}\sum_{t=0}^{T-1} \delta_{X_t}. \tag{1}$$

This random probability measure records the proportion of time spent by the trajectory in each measurable set. An important question in the analysis of Markov chains (and, more broadly, of dynamical systems) is whether there exists a probability measure $\pi$ such that $\widehat{\mu}_T \to \pi$ as $T \to \infty$. Such a limit, when it exists, describes the statistical behavior of trajectories over long times (Eckmann and Ruelle, 1985; Young, 2002). For example, if $(X_t)_{t\geq 0}$ are independent random variables on a separable metric space $\mathfrak{X}$ with common law $\pi$, the empirical law of large numbers gives, almost surely,

$$\widehat{\mu}_T \rightharpoonup \pi, \qquad \text{or, equivalently,} \qquad \int f\, d\widehat{\mu}_T = \frac{1}{T}\sum_{t=0}^{T-1} f(X_t) \to \int f\, d\pi \tag{2}$$

for every bounded continuous test function $f$. For general Markov chains, however, convergence to a deterministic limiting measure is not automatic. For classical conditions ensuring almost-sure convergence of $\widehat{\mu}_T$ as in (2), see, e.g., Meyn and Tweedie (2012).

To quantify the discrepancy between $\widehat{\mu}_T$ and $\pi$, we need to choose a distance on the space of probability measures. A flexible framework is provided by *integral probability metrics* (IPMs),

$$d_{\mathcal{F}}(\mu,\nu) := \sup_{f\in\mathcal{F}} \left| \int f\, d\mu - \int f\, d\nu \right|, \tag{3}$$

where $\mathscr{F}$ is a class of measurable test functions integrable with respect to the measures under consideration. The choice of $\mathscr{F}$ determines which differences between probability measures are detected, and hence the notion of convergence being studied. For example, the class of measurable functions satisfying $\|f\|_\infty \le 1$ gives the total variation norm, while the class of 1-Lipschitz functions gives the Wasserstein distance $W_1$ on probability measures with finite first moments. Further examples and general properties of IPMs can be found in Müller (1997) and Rachev et al. (2013).

Our aim is to obtain non-asymptotic bounds on $d_{\mathscr{F}}(\widehat{\mu}_T, \pi)$, for suitable choices of $\mathscr{F}$, with explicit dependence on the trajectory length $T$. Such bounds are useful in sampling, where they quantify the error incurred when approximating a target distribution by a finite trajectory of a Markov chain (Roberts and Rosenthal, 2004; Joulin and Ollivier, 2010). Related questions also arise in the analysis of stochastic optimization algorithms through their interpretation as Markov chains (Dieuleveut et al., 2020; Yu et al., 2021).

Non-asymptotic bounds have been studied in several probability metrics. For uniformly ergodic Markov chains on countable state spaces, bounds in total variation follow from the concentration and moment inequalities of Paulin (2015). Bounds in $W_1$ under suitable contractivity assumptions have been obtained by Boissard (2011); Kloeckner (2020); Riekert (2022). For $W_p$, Fournier and Guillin (2015) establish moment bounds for empirical measures of independent samples and Markov chains. Related bounds for occupation measures of continuous-time ergodic Markov processes are obtained by Schilling et al. (2025) under Wasserstein contractivity or functional inequalities, together with moment assumptions on the limiting measure.

In this work, we consider IPMs obtained from reproducing kernel Hilbert space (RKHS), known as *maximum mean discrepancies* (MMD). Let $k : \mathcal{X} \times \mathcal{X} \to \mathbb{R}$ be a measurable positive-definite kernel with separable RKHS $\mathscr{H}$, and let $\phi(x) := k(x, \cdot)$ be its canonical feature map. Taking $\mathscr{F}$ to be the unit ball of $\mathscr{H}$ in (3), Hilbert-space duality gives

$$\mathrm{MMD}_\phi(\mu, \nu) = \left\| \int \phi(x)\, \mu(dx) - \int \phi(x)\, \nu(dx) \right\|_{\mathscr{H}}, \tag{4}$$

whenever the two Bochner integrals exist; see, e.g., Smola et al. (2007); Gretton et al. (2012). The same construction applies to any measurable feature map $\phi : \mathcal{X} \to \mathscr{H}$ into a separable Hilbert space, subject to the same integrability requirement. The resulting discrepancy is the MMD associated with the kernel $k(x, y) = \langle \phi(x), \phi(y) \rangle_{\mathscr{H}}$. We develop this feature-map formulation in Section 3.

When the feature map is uniformly bounded, that is, $\sup_{x \in \mathcal{X}} \|\phi(x)\|_{\mathscr{H}} < \infty$, the discrepancy is well defined for all probability measures, including empirical occupation measures. For independent observations with common law $\pi$, $\mathrm{MMD}_\phi(\widehat{\mu}_T, \pi)$ admits bounds of order $T^{-1/2}$, both in expectation and at each fixed confidence level (Gretton et al., 2012). For dependent observations, Chérief-Abdellatif and Alquier (2022) obtain high-probability bounds under weak-dependence assumptions in their analysis of minimum-MMD estimators. More recently, Wolfer and Alquier (2025) derive such bounds for stationary $\varphi$- and $\beta$-mixing sequences.

**Contributions.** Our first contribution is a non-asymptotic high-probability bound for empirical occupation measures of Markov chains on a general state space. Assume that the chain is stationary, has an absolute $L^2(\pi)$ spectral gap $1 - \rho > 0$, and satisfies the following uniform geometric relaxation condition:

$$\sup_{x \in \mathcal{X}} \mathrm{MMD}_\phi(P^n(x, \cdot), \pi) \le C\theta^n, \qquad n \ge 0, \tag{5}$$

for some $C < \infty$ and $\theta \in [0, 1)$. Then, for every $T \ge 1$ and $\delta \in (0, 1)$, with probability at least $1 - \delta$,

$$\mathrm{MMD}_\phi(\widehat{\mu}_T, \pi) \lesssim \|\phi\|_\pi \sqrt{\frac{\tau_{\mathrm{rel}} \log(4/\delta)}{T}} + \frac{\tau_\theta \sqrt{\tau_{\mathrm{rel}}} \log(4/\delta)}{T}, \tag{6}$$